

New Attempt

- Due Nov 27, 2024 by 2:59am
- Points 100
- Submitting a text entry box or a file upload

Individual Laser Cutting Project

Overview

This individual assignment gives you the chance to use the Engineering Design Process and apply your AutoCAD skills to improve the functionality of your robot by designing and laser cutting one or two components. Keep in mind, you're not just creating a simple part—you're developing a functional piece that showcases your CAD expertise and problem-solving skills.

Happy designing!! 🙌



Learning Objectives

1. **AutoCAD Design Proficiency:** Develop and demonstrate skills in designing a part in AutoCAD that can be laser cut.
2. **Real-Life Engineering Challenge:** Tackle a practical engineering design challenge using the engineering design process, simulating scenarios encountered in industry or research while adhering to specific design constraints.
3. **Understanding laser cutting:** Learn how a laser cutting machine works and observe it in action as it brings your design to life.



Assignment

Problem Statement

After testing your robot's performance in Robot Project Challenge 1, identify a specific issue affecting its maneuverability or overall functionality. Using the Engineering Design Process, develop solutions and design 1-2 custom parts in AutoCAD that can be laser-cut from acrylic and integrated into the existing chassis to enhance your robot's performance. This is an individual assignment, and each student is responsible for contributing fully to the design process and final submission. 🎉🎉🎉

The following are the project deliverables:

Individual Write-up Submitted to Canvas by 11/26

Using the Engineering Design Process, complete and document the following steps in 4 pages Max:

1. **Define the Problem:** Identify a specific improvement needed in your robot's maneuverability or functionality, and define the problem.
2. **Develop Solutions:** Generate multiple design concepts and include photos of your hand-drawn sketches.
3. **Evaluate Solutions:** Use KTDA (Kepner-Tregoe Decision Analysis) to select the best solution and include the table in your submission.
4. **CAD Design:** draw your selected design in AutoCAD using the provided **[AutoCAD laser cutting template \(https://northeastern.instructure.com/courses/189179/files/31022864?wrap=1\)](https://northeastern.instructure.com/courses/189179/files/31022864?wrap=1)**  (**https://northeastern.instructure.com/courses/189179/files/31022864/download?download_frd=1**) . Include a screenshot of your final design in AutoCAD in your write up.
5. **Prepare for Laser Cutting:** Follow the laser cutting instructions and project guidelines below to prepare your file and submit for laser cutting.
6. **Test and Reflect:** Include photo(s) of the laser-cut parts individually and attached to your robot for reference. Evaluate their impact on performance through testing, and then reflect on their effectiveness in solving the identified problem.

Final Submission: Submit your completed write-up in Pdf with the CAD files to Canvas by Nov. 26 Midnight!

CAD and Laser Cutting Files (Laser cutting file submitted to Prof. Keyvani by 11/22)

1. **Material:** Your design will be cut from 1/8-inch (.118") thick acrylic for free.
2. **Template:** Please use the this **[AutoCAD template \(https://northeastern.instructure.com/courses/189179/files/31022884?wrap=1\)](https://northeastern.instructure.com/courses/189179/files/31022884?wrap=1)**  (**https://northeastern.instructure.com/courses/189179/files/31022884/download?download_frd=1**)_for your design to ensure it can be cut. This template is an AutoCAD .dwt

file that you can save to your computer or Vlab. To start, open AutoCAD, select 'New,' and choose this template file.

3. **Draw to scale using the layers provided. Note that the laser cutter will remove 0.05 inch of material from each cut side.**
4. **Add Details:** Please be sure to add your name to your part using the text command. Note that text does not directly transfer to the laser cutting software; to make it visible, you'll need to explode the text using the TXTEXP command. *Important:* AutoCAD for Mac does not support the TXTEXP function needed to explode text. To complete this step, please use the Windows version of AutoCAD (on Vlab for example). Follow the steps below to explode text on Windows version of AutoCAD:
 - Select all of your text
 - In the command line at the bottom, type in "TXTEXP" and click enter. Note: TXTEXP is for exploding text only, the command "EXPLODE" will not work for exploding text.
 - All text will now appear as empty block letters. This means your text is correctly exploded. Please use the etch layer (blue layer) for your text.
5. **Saving your file for submission to Canvas:** Save your design as .dwg and .pdf with a descriptive file name and submit to Canvas. *Example file name:*
smith_acrylic_cut_Cornerstone.dwg.
6. **Exporting your file for Laser Cutting:** Save your .dwg file as .dxf to be laser cut.
 - Click File and then Save As
 - In the Save Drawing As dialog box, select a DXF format in the Files of type box
 - Select a folder and enter a file name
 - Click Save
7. **Cutting your design:** The Makerspace will cut designs from Saturday, 11/9, through Tuesday, 11/26. To have your file included in an upcoming batch, please send your laser-cutting file (.dxf) via MS Teams to Prof. Keyvani by 2:30 PM on Friday (11/8, 11/15, or 11/22). Prof. Keyvani will compile all files submitted by the deadline and deliver them to the Makerspace staff for cutting. Late submissions will be included in the following week's batch, as the Makerspace staff cannot accept individually delivered files. **Please note:** November 22 is the final deadline for submitting files for laser cutting.

Things to Consider

- Use the correct layer: **Red - Cut:** *A high powered and slow moving beam that cuts through material.* **Blue - Outline (Etch):** *A low powered and fast-moving beam that creates a shallow cut on top of the material.*
- **Note that the laser cutter will remove 0.05 inch of material from each cut side. A 1x1 inch cut on acrylic ends up measuring approximately 0.99 x 0.99 inches**
- In your drawing file, **please arrange all pieces as close together as possible**, removing any unnecessary gaps, to optimize material usage effectively.
- You don't need to fill the entire material area shown in the template.
- Make sure there are no intersecting lines.

- Very small text or detailed fonts may not translate when laser cutting.

Final Submission

1. Submit your write up --- 50 pts
2. AutoCAD drawing file (.dwg) --- 40 pts
- 3- A pdf version of the drawing file. (pdf) --- 10 pts

AutoCAD laser Cutting							
Criteria	Ratings						Pts
Write up 5 steps (10 pts each). Problem Solution KTDA Screenshot of CAD drawing and photo of laser cut piece testing and evaluation	50 pts Full Marks	40 pts Missing one	30 pts Missing two	20 pts Missing three	10 pts Missing four	0 pts No Marks	50 pts
AutoCAD drawing/pdf Used template Has cut line Has name/initials has pdf matches with the lasercut part	50 pts Full Marks	40 pts Missing one	30 pts Missing two	20 pts Missing three	10 pts Missing four	0 pts No Marks	50 pts
Total Points: 100							